

from different families and perhaps with different architectures. Other directions for future work include applications across different domains and modalities, investigating variants with simultaneous multi-token patching or multi-layer patching to mitigate the risks of placeholder contamination, and presenting recipes for task-specific and task-agnostic Patchscopes.

Acknowledgements

We thank Amir Globerson for feedback on writing and presentation of results. We thank Ardavan Saeedi, Martin Wattenberg, Ellie Pavlick, the AI Explorables team³ at Google Research, and Jasmijn Bastings for their helpful comments.

Impact Statement

Societal Impact This paper presents a new framework for interpreting hidden representations of large language models. Interpretability methods in general can be used to investigate models’ reliability and safety prior to deployment. We hope that Patchscopes framework facilitates progress in this area with the introduction of inspection tools that are more expressive, robust across layers, and do not require training data.

Limitations While our proposed framework is not limited to any particular architecture or domain, the experimental evidence provided in this paper focuses on autoregressive Transformer-based language models, and future work is needed to verify its effectiveness in other setups.

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